📊 Nivesh AI — Technical Deep Dive & Results

🧠 Core Intelligence Engine

Nivesh AI is not just a dashboard — it is a **signal detection engine**.

We implemented:

- RSI (Relative Strength Index) for momentum detection
- Price movement analysis for breakout/breakdown signals
- Rule-based AI layer to convert signals into human-readable insights

This allows the system to move from **raw data → decision-ready output**.

⚙️ Signal Logic

Breakout Detection

- If price increases > 3% from previous close
  → Mark as bullish breakout

Breakdown Detection

- If price drops > 3%
  → Mark as bearish signal

RSI Interpretation

- RSI > 60 → strong momentum
- RSI < 40 → weak momentum
- RSI ~ 50 → neutral

🧩 System Components

1. Data Layer

- Fetches stock data using yfinance
- Handles real-time-like market data

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2. Processing Layer

- Calculates indicators (RSI, MACD)

- Detects signals

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3. AI Layer

- Converts signals into insights
- Generates recommendations

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4. API Layer (FastAPI)

- /analyze → stock analysis
- /scan → opportunity radar
- /chat → AI responses
- /video → market summary

📊 Sample Output

Example: RELIANCE.NS

- Price: ₹1348
- Signal: Breakdown
- RSI: 50
- Recommendation: SELL

AI Insight:

"Reliance is showing bearish movement with selling pressure increasing."

📈 Key Results

- Reduced decision complexity for users
- Enabled instant stock analysis
- Automated opportunity detection
- Simplified financial insights

🚀 Scalability

The architecture supports:

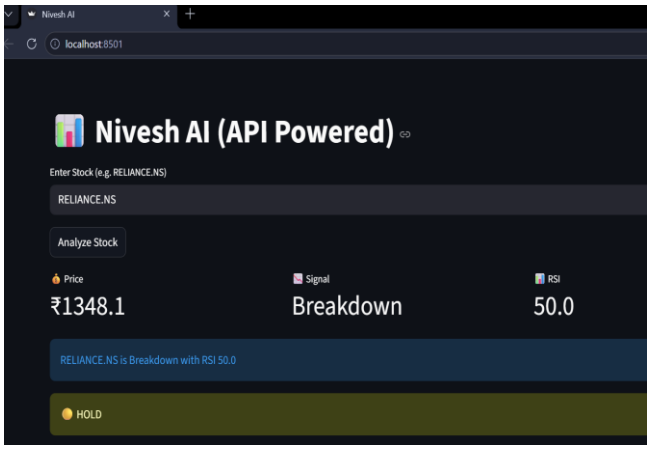
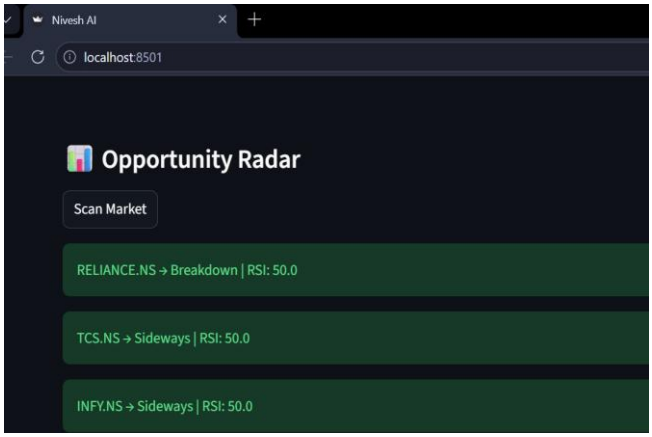
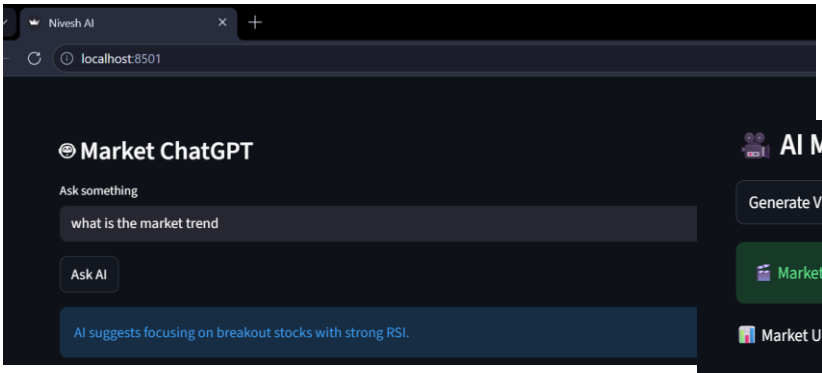
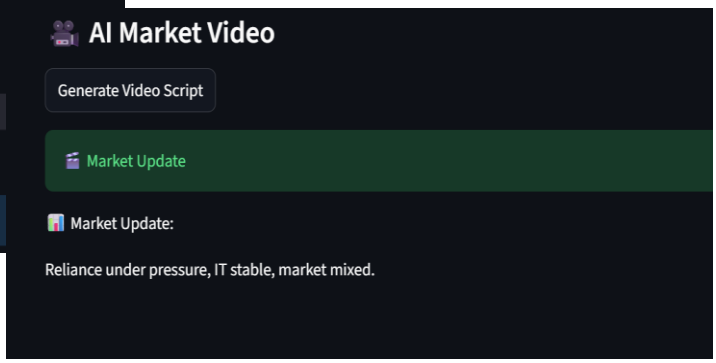
- Real-time data APIs (NSE, Zerodha)
- AI/LLM integration (ChatGPT-style reasoning)
- Portfolio-aware recommendations

🧠 Key Innovation

Traditional platforms show charts.
Nivesh AI explains decisions.

🏁 Final Insight

The real value of Nivesh AI lies in bridging the gap between:

👉 Data → Understanding → Action

This is what transforms a retail investor into a smart investor.